

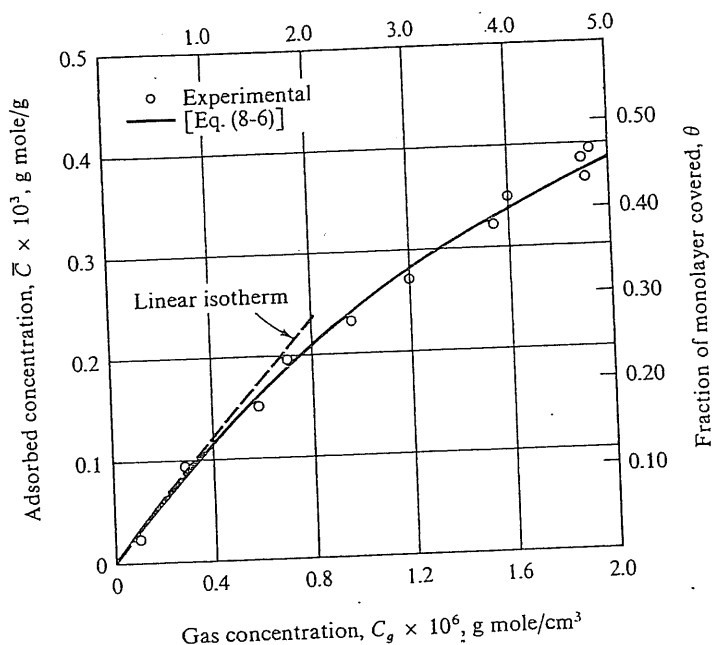


Fig. 8-2 Adsorption-equilibrium data for *n*-butane on silica gel (surface area 832 m²/g) at 50°C

Chemisorption data often do not fit Eq. (8-6). However, the basic concepts on which the Langmuir isotherm is based, the ideas of a dynamic equilibrium between rates of adsorption and desorption and a finite adsorption time, are sound and of great value in developing the kinetics of fluid-solid catalytic reactions. Equations (8-4) to (8-6) form the basis for the rate equations presented in Chap. 9.

PHYSICAL PROPERTIES OF CATALYSTS

The surface area of a solid has a pronounced effect on the amount of gas adsorbed and on its activity as a catalyst. For example, if a sample of fresh Raney nickel, which is highly porous and has a large surface, is held in the hand, the heat due to adsorption of oxygen can be felt immediately. No heat is apparent in the same mass of nonporous nickel. This relationship between surface area and extent of adsorption has led to the development of highly porous materials with areas as high as 1,500-m²/g. Sometimes the catalytic material itself can be prepared in a form with large surface area. When this is not possible, materials which can be so prepared may

be used as a carrier or support on which the catalytic substance is dispersed. Silica gel and alumina are widely used as supports.

The dependence of rates of adsorption and catalytic reactions on surface makes it imperative to have a reliable method of measuring surface area. Otherwise it would not be possible to compare different catalysts (whose areas are different) to ascertain the intrinsic activity per unit surface. For surface areas in the range of hundreds of square meters per gram a porous material with equivalent cylindrical pore radii (see Sec. 8-5) in the range of 10 to 100 Å is needed. The following example shows that such areas are not possible with nonporous particles of the size which can be economically manufactured.

Example 8-1 Spray drying and other procedures for manufacturing small particles can produce particles as small as 2 to 5 microns. Calculate the external surface area of nonporous spherical particles of 2 microns diameter. What size particles would be necessary if the external surface is to be 100 m²/g? The density of the particles is 2.0 g/cm³.

Solution The external surface area per unit volume of a spherical particle of diameter d_p is

$$\frac{\pi d_p^2}{\pi d_p^3/6} = \frac{6}{d_p}$$

If the particle density is ρ_p , the surface area, per gram of particles, would be

$$S_g = \frac{6}{\rho_p d_p}$$

For $d_p = 2$ microns (2×10^{-4} cm) and $\rho_p = 2.0$ g/cm³

$$S_g = \frac{6}{2(2 \times 10^{-4})} = 1.5 \times 10^4 \text{ cm}^2/\text{g}$$

or

$$S_g = 1.5 \text{ m}^2/\text{g}$$

This is about the largest surface area to be expected for nonporous particles. If a surface of 100 m²/g were required, the spherical particles would have a diameter of

$$d_p = \frac{6}{\rho_p S_g} = \frac{6}{2.0(100 \times 10^4)} = 0.02 \times 10^{-4} \text{ cm}$$

or

$$d_p = 0.02 \text{ micron}$$

Particles as small as this cannot, at present, be produced on a commercial scale. It may be noted that the smaller particles found in a fluidized-bed reactor are retained on 400 mesh size, which has a sieve opening of 37 microns.

When the major catalytic surface is in the interior of a solid particle, the resistance to transport of mass and energy from the external surface to the interior can have a significant effect on the global rate of reaction. Quantitative treatment of this problem is the objective in Chap. 11. It is sufficient here to note that this treatment rests on a geometric model for the extent and distribution of void spaces within the complex porous structure of the particle. It would be best to know the size and shape of each void space in the particle. In the absence of this information the parameters in the model should be evaluated from reliable and readily obtainable geometric properties. In addition to the surface area, three other properties fall into this classification: void volume, the density of the solid material in the particle, and the distribution of void volume according to void size (pore-volume distribution). The methods of measurement of these four properties are considered in Secs. 8-5 to 8-7.

8-5 Determination of Surface Area

The standard method for measuring catalyst areas is based on the physical adsorption of a gas on the solid surface. Usually the amount of nitrogen adsorbed at equilibrium at the normal boiling point (-195.8°C) is measured over a range of nitrogen pressures below 1 atm. Under these conditions several layers of molecules may be adsorbed on top of each other on the surface. The amount adsorbed when one molecular layer is attained must be identified in order to determine the area. The historical steps in the development of the Brunauer-Emmett-Teller method¹ are clearly explained by Emmett.² There may be some uncertainty as to whether the values given by this method correspond exactly to the surface area. However, this is relatively unimportant, since the procedure is standardized and the results are reproducible. It should be noted that the surface area so measured may not be the area effective for catalysis. For example, only certain parts of the surface, the active centers, may be active for chemisorption, while nitrogen may be physically adsorbed on much more of the surface. Also, when the catalyst is dispersed on a large-area support, only part of the support area may be covered by catalytically active atoms. For example, a nickel-on-kieselguhr catalyst was found to have a surface of $205\text{ m}^2/\text{g}$ as measured by nitrogen adsorption.³ To determine the area covered by nickel atoms, hydrogen was chemisorbed on the catalyst at 25°C . From the amount of hydrogen chemisorbed, the surface area of nickel atoms was calculated to be about $40\text{ m}^2/\text{g}$. It would be most useful to know surface areas for

¹S. Brunauer, P. H. Emmett, and E. Teller, *J. Am. Chem. Soc.*, **60**, 309 (1938).

²P. H. Emmett (ed.), "Catalysis," vol. I, chap. 2, Reinhold Publishing Corporation, New York, 1954.

³G. Padberg and J. M. Smith, *J. Catalysis*, **12**, 111 (1968).

chemisorption of the reactant at reaction conditions. However, this would require measurement of relatively small amounts of chemisorption at different, and often troublesome, conditions (high temperature and/or pressure), for each reaction system. In contrast, nitrogen can be adsorbed easily and rapidly in a routine fashion with standard equipment.

In the classical method of determining surface area an all-glass apparatus is used to measure the volume of gas adsorbed on a sample of the solid material.¹ The apparatus operates at a low pressure which can be varied from near zero up to about 1 atm. The operating temperature is in the range of the normal boiling point. The data obtained are gas volumes at a series of pressures in the adsorption chamber. The observed volumes are normally corrected to cubic centimeters at 0°C and 1 atm (standard temperature and pressure) and plotted against the pressure in millimeters, or as the ratio of the pressure to the vapor pressure at the operating temperature. Typical results from Brunauer and Emmett's work² are shown in Fig. 8-3 for the adsorption of several gases on a 0.606-g sample of silica gel. To simplify the classical experimental procedure a flow method has been developed in which a mixture of helium and the gas to be adsorbed is passed continuously over the sample of solid.³ The operating total pressure is constant, and the partial pressure of adsorbable gas is varied by changing the composition of the mixture. The procedure⁴ is to pass a mixture of known composition over the sample until equilibrium is reached, that is, until the solid has adsorbed an amount of adsorbable component corresponding to equilibrium at its partial pressure in the mixture. Then the gas is desorbed by heating the sample while a stream of pure helium flows over it. The amount desorbed is measured with a thermal-conductivity cell or other detector. This gives one point on an isotherm, such as shown in Fig. 8-3. Then the process is repeated at successively different compositions of the mixture until the whole isotherm is obtained.

The curves in Fig. 8-3 are similar to the extent that at low pressures they rise more or less steeply and then flatten out for a linear section at intermediate pressures. After careful analysis of much data it was concluded that the lower part of the linear region corresponded to complete monomolecular adsorption. If this point could be located with precision, the

¹For a complete description of apparatus and techniques see L. G. Joyner, "Scientific and Industrial Glass Blowing and Laboratory Techniques," Instruments Publishing Company, Pittsburgh, 1949; see also S. Brunauer, "The Adsorption of Gases and Vapors," vol. 1, Princeton University Press, Princeton, N.J., 1943.

²S. Brunauer and P. H. Emmett, *J. Am. Chem. Soc.*, **59**, 2682 (1937).

³F. M. Nelson and F. T. Eggertsen, *Anal. Chem.*, **30**, 1387 (1958).

⁴A description of the operating procedure and the data obtained are given by S. Masamune and J. M. Smith [*AIChE J.*, **10**, 246 (1964)] for the adsorption of nitrogen on Vycor (porous glass).

volume of one monomolecular layer of gas, v_m , could then be read from the curve and the surface area evaluated. The Brunauer-Emmett-Teller method locates this point from an equation obtained by extending the Langmuir isotherm to apply to multilayer adsorption. The development is briefly summarized as follows: Equation (8-3) can be rearranged to the form

$$\frac{p}{v} = \frac{1}{Kv_m} + \frac{p}{v_m} \quad (8-8)$$

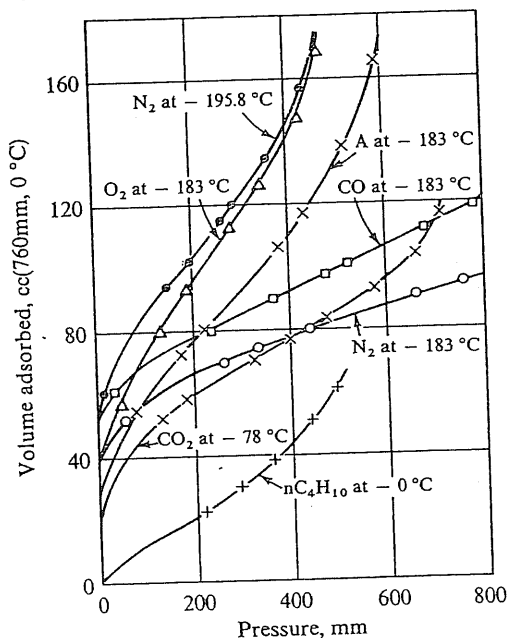
Brunauer, Emmett, and Teller adapted this equation for multilayer adsorption and arrived at the result

$$\frac{p}{v(p_0 - p)} = \frac{1}{v_m c} + \frac{(c - 1)p}{cv_m p_0} \quad (8-9)$$

where p_0 is the saturation or vapor pressure and c is a constant for the particular temperature and gas-solid system.

According to Eq. (8-9), a plot of $p/v(p_0 - p)$ vs p/p_0 should give a straight line. The data of Fig. 8-3 are replotted in this fashion in Fig. 8-4.

Fig. 8-3 Adsorption isotherms for various gases on a 0.606-g sample of silica gel [by permission from P. H. Emmett (ed.), "Catalysis," vol. I, Reinhold Publishing Corporation, New York, 1954]



catalytic reaction $\text{H}_2 \rightarrow 2\text{H}$ on a hot tungsten filament, for which both rate of reaction and ΔH_a -vs- θ data are available.¹

9-3 Rates of Adsorption and Desorption

In this section equations are presented for the rates of adsorption, surface reaction, and desorption. In Sec. 9-4 these equations will be combined to give expressions for the rate in terms of fluid concentrations; that is, concentrations on the surface (denoted by $\bar{C}$) will be eliminated.

Adsorption The net rate of adsorption of a component A is given by the difference between Eqs. (8-4) and (8-5), written as²

$$r_a = k_a C_A (\bar{C}_m - \bar{C}) - k_a \bar{C}_A = k_a \left[C_A (\bar{C}_m - \bar{C}) - \frac{1}{K_A} \bar{C}_A \right]$$

where $\bar{C}_m - \bar{C}$ represents the concentration of vacant sites, $\bar{C}_v$. If only A were adsorbed, then $\bar{C}_m - \bar{C}$ would equal $\bar{C}_m - \bar{C}_A$. However, other components of the reaction may be adsorbed, so that it is necessary to write $\bar{C}_m - \bar{C} = \bar{C}_v$, where $\bar{C}_v$ is expressed as the concentration of vacant sites per unit mass of catalyst, or

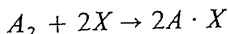
$$r_a = k_a \left(C_A \bar{C}_v - \frac{1}{K_A} \bar{C}_A \right) \quad (9-14)$$

In this equation C_A is the concentration of A in the gas phase at the catalyst surface. If the resistance to adsorption is negligible with respect to other steps in the overall conversion process, the concentration of A on the catalyst surface is in equilibrium with the concentration of A in the gas phase. The net rate of adsorption, from Eq. (9-14), approaches zero, and the equilibrium concentration of A is given by the expression

$$(\bar{C}_A)_{\text{eq}} = K_A C_A \bar{C}_v \quad (9-15)$$

where K_A designates the adsorption equilibrium constant for A .

Note that this result would reduce to the Langmuir isotherm, Eq. (8-6), if only A were adsorbed. Equations (9-14) and (9-15) are applicable when A occupies one site. Often in chemisorption a diatomic molecule, such as oxygen, will dissociate upon adsorption with each atom occupying one site. Formally, dissociative adsorption may be written



¹ M. Boudart, *Ind. Chim. Belge*, 23, 383 (1958).

² The gas phase concentration C_g is now written as C_A to denote component A . Also, the rate constant k_c is written as k_a to denote adsorption.

For this case the net rate of adsorption is

$$r_a = k_a \left(C_{A_2} \bar{C}_v^2 - \frac{1}{K_{A_2}} \bar{C}_A^2 \right) \quad (9-14')$$

At equilibrium the concentration of atomically adsorbed A is¹

$$(\bar{C}_A)_{eq} = K_{A_2}^{\frac{1}{2}} C_{A_2}^{\frac{1}{2}} \bar{C}_v \quad (9-15')$$

Surface Reaction The mechanism assumed for the surface process will depend on the nature of the reaction. Suppose that the overall reaction is of the type



An immediate question concerning the surface process is whether the reaction is between an adsorbed molecule of A and a gaseous molecule of B at the surface or between adsorbed molecules of both A and B on adjacent active centers. In the former case the process might be represented by the expression



If the concentration of adsorbed product C on the surface is $\bar{C}_C$, in moles per unit mass of catalyst (analogous to $\bar{C}_A$), the net rate of this step would be

$$r_s = k_s \bar{C}_A C_B - k'_s \bar{C}_C = k_s \left(\bar{C}_A C_B - \frac{1}{K_s} \bar{C}_C \right) \quad (9-18)$$

In this equation we suppose that the rate of the forward reaction is first order in A on the solid surface and first order in B in the gas phase. Similarly, the rate of the reverse process is first order in C on the surface.

If the mechanism is a reaction between adsorbed A and adsorbed B , the process may be represented by the expression



Here only those A molecules will react which are adsorbed on sites immediately adjacent to adsorbed B molecules. Hence the rate of the forward reaction should be proportional to the concentration of the pairs of adjacent sites occupied by A and B . The concentration of these pairs will be equal to $\bar{C}_A$ multiplied by the fraction of the adjacent sites occupied by B molecules.

¹If A_2 is the only adsorbable gas, then $\bar{C}_v = \bar{C}_m - \bar{C}_A$, and Eq. (9-15') becomes

$$(\bar{C}_A)_{eq} = \frac{K_{A_2}^{\frac{1}{2}} C_{A_2}^{\frac{1}{2}} \bar{C}_m}{1 + K_{A_2}^{\frac{1}{2}} C_{A_2}^{\frac{1}{2}}} \quad \text{or} \quad \bar{\theta}_A = \frac{K_{A_2}^{\frac{1}{2}} C_{A_2}^{\frac{1}{2}}}{1 + K_{A_2}^{\frac{1}{2}} C_{A_2}^{\frac{1}{2}}}$$

in comparison with Eqs. (8-6) and (8-7) for molecularly adsorbed A_2 .

This fraction is proportional to the fraction of the total surface occupied by B molecules, i.e., to θ_B .¹ If $\bar{C}_m$ is defined as the molal concentration of total sites, then $\theta_B = \bar{C}_B/\bar{C}_m$. The rate of the forward reaction, according to the mechanism of Eq. (9-19), will be

$$r = k_s \bar{C}_A \frac{\bar{C}_B}{\bar{C}_m}$$

The reverse rate is proportional to the pairs of centers formed by adsorbed product C molecules and adjacent vacant centers,

$$r' = k'_s \bar{C}_C \frac{\bar{C}_v}{\bar{C}_m}$$

Combining these two expressions gives the net surface rate by the mechanism of Eq. (9-19),

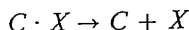
$$r_s = \frac{1}{\bar{C}_m} (k_s \bar{C}_A \bar{C}_B - k'_s \bar{C}_C \bar{C}_v) = \frac{k_s}{\bar{C}_m} \left(\bar{C}_A \bar{C}_B - \frac{1}{K_s} \bar{C}_C \bar{C}_v \right) \quad (9-20)$$

If the surface step has a negligible resistance with respect to the others, the process would occur at equilibrium, and Eq. (9-18) or Eq. (9-20) could be used to relate the concentrations of the reactants and products on the catalyst surface. For example, if the chosen mechanism is Eq. (9-19), the concentration of product C is given by Eq. (9-20) with $r_s = 0$; that is,

$$K_s = \left(\frac{\bar{C}_v \bar{C}_C}{\bar{C}_A \bar{C}_B} \right)_{\text{eq}} \quad (9-21)$$

where K_s is the equilibrium constant for the surface reaction.

Desorption The mechanism for the desorption of the product C may be represented by the expression



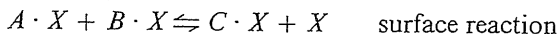
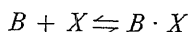
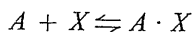
The rate of desorption will be analogous to Eq. (9-14) for the adsorption of A ,

$$r_d = k'_d \bar{C}_C - k_d C_C \bar{C}_v = -k_d \left(C_C \bar{C}_v - \frac{1}{K_c} \bar{C}_C \right) \quad (9-22)$$

¹This is strictly correct only if the fraction of the surface occupied by A molecules is small. It would be more accurate to postulate that the fraction of the adjacent centers occupied by B is equal to $\theta_B/(1 - \theta_A)$. For small values of θ_A the two results are nearly the same.

9-4 Rate Equations in Terms of Concentrations in the Fluid Phase at the Catalyst Surface

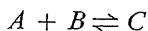
At steady state the rates of adsorption r_a , surface reaction r_s , and desorption r_d are equal. To express the rate solely in terms of fluid concentrations, the adsorbed concentrations $\bar{C}_A$, $\bar{C}_B$, $\bar{C}_C$, and $\bar{C}_v$ must be eliminated from Eqs. (9-15) to (9-22). In principle, this can be done for any reaction, but the resultant rate equation involves all the rate constants k_i and the equilibrium constants K_i . Normally neither type of constant can be evaluated independently.¹ Both must be determined from measurements of the rate of conversion from fluid reactants to fluid products. However, there are far too many constants, even for simple reactions, to obtain meaningful values from such overall rate data. The problem can be eased, with some confidence from experimental data, by supposing that one step in the overall reaction controls the rate. Then the other two steps occur at near-equilibrium conditions. This greatly simplifies the rate expression and reduces the number of rate and equilibrium constants that must be determined from experiment. To illustrate the procedure equations for the rate will be developed, for various controlling steps, for the reaction system



adsorption

surface reaction

desorption



overall reaction

Surface Reaction Controlling The concentrations $\bar{C}_A$, $\bar{C}_B$, and $\bar{C}_C$ will be those corresponding to equilibrium for the adsorption and desorption steps. Equation (9-15) gives the equilibrium value for $\bar{C}_A$. Similar results for $\bar{C}_B$ and $\bar{C}_C$ are

$$(\bar{C}_B)_{\text{eq}} = K_B C_B \bar{C}_v \quad (9-23)$$

$$(\bar{C}_C)_{\text{eq}} = K_C C_C \bar{C}_v \quad (9-24)$$

Substituting these results in Eq. (9-20) for the surface rate gives

$$r = \frac{k_s}{\bar{C}_m} \left(K_A K_B C_A C_B \bar{C}_v^2 - \frac{K_C}{K_s} C_C \bar{C}_v^2 \right) \quad (9-25)$$

¹It might be supposed that rate and equilibrium constants for adsorption or desorption could be established from pure-component adsorption data on the components involved. However, such results rarely agree with the values of the constants determined from rate data for the reaction. Interaction effects between components, other inadequacies of the Langmuir theory, and the assumption of a single controlling step, explain the deviation.

The concentration of vacant sites can be expressed in terms of the total concentration of sites $\bar{C}_m$,

$$\bar{C}_m = \bar{C}_v + \bar{C}_A + \bar{C}_B + \bar{C}_C \quad (9-26)$$

Since $\bar{C}_A$ (or $\bar{C}_B$ and $\bar{C}_C$) corresponds to the equilibrium value for adsorption, Eqs. (9-15), (9-23), and (9-24) can be combined with Eq. (9-26) to yield

$$\bar{C}_v = \frac{\bar{C}_m}{1 + K_A C_A + K_B C_B + K_C C_C} \quad (9-27)$$

Now Eqs. (9-25) and (9-27) can be combined to give a relatively simple expression for the rate in terms of fluid concentrations,

$$r = k_s \bar{C}_m \frac{K_A K_B C_A C_B - (K_C/K_s) C_C}{(1 + K_A C_A + K_B C_B + K_C C_C)^2} \quad (9-28)$$

This result can be further reduced by using the relationship between the several equilibrium constants K_A , K_B , K_C , and K_s . If the equilibrium constant for the overall reaction is denoted by K , then

$$K = \left(\frac{C_C}{C_A C_B} \right)_{\text{eq}} \quad (9-29)$$

This is the conventional K for a homogeneous reaction computed from thermodynamic data, as outlined in Chap. 2. It may be related to the adsorption and surface-reaction equilibrium constants by the equilibrium equations for each of these processes, Eqs. (9-15), (9-23), and (9-24). Thus

$$K = \frac{\bar{C}_C/K_C \bar{C}_v}{(\bar{C}_A/K_A \bar{C}_v)(\bar{C}_B/K_B \bar{C}_v)} = \frac{K_A K_B}{K_C} \left(\frac{\bar{C}_v \bar{C}_C}{\bar{C}_A \bar{C}_B} \right)_{\text{eq}} \quad (9-30)$$

According to Eq. (9-21), the last group of surface concentrations is K_s , and so

$$K = \frac{K_A K_B}{K_C} K_s \quad (9-31)$$

Substituting this relationship in Eq. (9-28) gives the final expression for the rate in terms of fluid concentrations,

$$r = k_s \bar{C}_m K_A K_B \frac{C_A C_B - (1/K) C_C}{(1 + K_A C_A + K_B C_B + K_C C_C)^2} \quad (9-32)$$

It is well at this point to review the chief premises involved in Eq. (9-32):

1. It is supposed that the surface reaction controls the rate of the three steps.
2. The equation applies to the simple reaction $A + B \rightarrow C$. Furthermore, it is assumed that the mechanism of the surface reaction involves the

combination of an adsorbed molecule of A and an adsorbed molecule of B .

3. The concepts of adsorption and kinetics rest on the Langmuir-Hinshelwood theories referred to earlier.

If the adsorption is weak for all components, the denominator of Eq. (9-32) approaches unity, and the rate expression reduces to the homogeneous form

$$r = k_s \bar{C}_m K_A K_B (C_A C_B - \frac{1}{K} C_C)$$

or

$$r = k(C_A C_B - \frac{1}{K} C_C) \quad (9-33)$$

The decomposition of formic acid on several catalytic surfaces follows such behavior.¹ Here there is only one reactant, so that the rate becomes first order in concentration of formic acid.

If the product of a reaction is strongly adsorbed and the reactant adsorption is weak, the term $K_C C_C$ is much larger than all others in the denominator. If the reaction is also irreversible, Eq. (9-32) becomes

$$r = k_s \bar{C}_m K_A K_B \frac{C_A C_B}{(K_C C_C)^2}$$

Suppose there was only one reactant; that is, the reaction was of the form $A \rightarrow C$. The denominator term would be raised to a power of unity instead of being squared, so that

$$r = k_s \bar{C}_m K_A \frac{C_A}{K_C C_C} \quad (9-34)$$

This result shows the retarding effect that a strongly adsorbed product can have on the rate. The decomposition of ammonia on a platinum wire has been found to follow this form of rate equation; i.e., hydrogen is strongly adsorbed and ammonia is only weakly adsorbed,² so that

$$r = k \frac{C_{\text{NH}_3}}{C_{\text{H}_2}}$$

¹C. N. Hinshelwood and B. Topley, *J. Chem. Soc.*, 123, 1014 (1923).

²C. N. Hinshelwood and R. E. Burk, *J. Chem. Soc.*, 127, 1105 (1925). Although there are two products in the decomposition, the nitrogen is only slightly adsorbed and does not appear in the rate equation.

Again consider a reaction of the form $A \rightarrow C$, but suppose that C is only weakly adsorbed, while A is very strongly adsorbed. Then the rate expression, in analogy with Eq. (9-34), would become zero order in A ; that is

$$r = k_s \bar{C}_m K_A \frac{C_A}{K_A C_A} = k \quad (9-35)$$

These special cases for the surface reaction controlling the rate all follow from the form of the Langmuir isotherm. Weak adsorption corresponds to small values of θ , and Eq. (9-7) shows that θ (or $\bar{C}_A$) is first order in p_A (or C_A), as predicted by Eq. (9-33). When the adsorption is very strong, the critical parameter is the concentration of vacant sites remaining for adsorption of reactant. Equation (9-9) shows that this is inversely proportional to the pressure of product (or $\bar{C}_C$), in agreement with Eq. (9-34).

Adsorption or Desorption Controlling Still retaining the simple reaction $A + B \rightleftharpoons C$, let us now suppose that the adsorption of A is the slow step. Then the adsorption of B , the surface reaction, and the desorption of C will occur at equilibrium. The rate can be formulated from the adsorption equation (9-14). The adsorbed concentration $\bar{C}_A$ in this expression is obtained from the equilibrium equations for the surface rate [Eq. (9-21)], adsorption of B [Eq. (9-23)], and desorption of C [Eq. (9-24)]. Thus

$$\bar{C}_A = \frac{\bar{C}_v \bar{C}_C}{K_s \bar{C}_B} = \frac{\bar{C}_v (K_C C_C \bar{C}_v)}{K_s (K_B C_B \bar{C}_v)} = \frac{\bar{C}_v K_C C_C}{K_s K_B C_B}$$

From the relationship of the several equilibrium constants, Eq. (9-31), the expression for $\bar{C}_A$ may be simplified to

$$\bar{C}_A = \frac{K_A \bar{C}_v C_C}{K C_B} \quad (9-36)$$

Substituting this value of $\bar{C}_A$ in the rate-controlling equation (9-14) gives

$$r = k_a \bar{C}_v \left(C_A - \frac{1}{K} \frac{C_C}{C_B} \right) \quad (9-37)$$

The expression for $\bar{C}_v$ can be formulated from Eq. (9-26), the equilibrium values of $\bar{C}_B$ and $\bar{C}_C$ from Eqs. (9-23) and (9-24), and $\bar{C}_A$ from Eq. (9-36). With this expression for $\bar{C}_v$ substituted in Eq. (9-37), the final rate equation for adsorption of A controlling the process is

$$r = \frac{k_a \bar{C}_m [C_A - (1/K)(C_C/C_B)]}{1 + K_B C_B + (K_A/K)(C_C/C_B) + K_C C_C} \quad (9-38)$$

If, instead of adsorption, the rate of desorption of product C controls the whole reaction, the expression for r should be formulated from Eq. (9-22). The adsorption and surface steps will occur at equilibrium conditions. Substituting the equilibrium values of $\bar{C}_C$ and $\bar{C}_v$ in Eq. (9-22) leads to the result²

$$r = k_d \bar{C}_m K \frac{C_A C_B - (1/K) C_C}{1 + K_A C_A + K_B C_B + K_C K C_A C_B} \quad (9-39)$$

9-5 Qualitative Analysis of Rate Equations

The procedure for developing rate expressions in terms of fluid properties according to the Langmuir concepts has been illustrated for a single reaction. Yang and Hougen¹ have considered a number of kinds of reactions and mechanisms and examined the results when adsorption, desorption, or surface reaction controls the rate. By dividing the final equation into a kinetic coefficient [for example, $k_d \bar{C}_m K$ in Eq. (9-39)], a driving force $[C_A C_B - (1/K) C_C]$, and an adsorption term $(1 + K_A C_A + K_B C_B + K_C K C_A C_B)$, they were able to prepare tables from which the rate equation for a specific situation could be quickly assembled.²

Equations such as (9-32), (9-38), and (9-39) will have value if they can be used to predict the rate over a wide range of conditions and hence be suitable for designing reactors. To be useful as working expressions the various constants (kinetics and equilibrium) must be given numerical values.³ It has not proved possible to obtain reliable K values from separate adsorption measurements. Hence all the constants must be determined from experimental kinetic data. This means, for example, that $k_d \bar{C}_m$, K_A , K_B , and K_C in Eq. (9-39) would be obtained from rate measurements. Since four-constant equations offer considerable flexibility, it is frequently possible to fit experimental data with equations based on several different mechanisms and assumptions regarding the step which is controlling. The advantage of this method of formulating rate reactions is the systematic way in which the pertinent parameters are introduced. It is a mechanized procedure which permits little flexibility and provides little insight into the actual mechanism of heterogeneous catalytic reactions. Progress in mecha-

¹K. H. Yang and O. A. Hougen, *Chem. Eng. Progr.*, **46**, 146 (1950).

²See also J. M. Thomas and W. J. Thomas, in "Introduction to the Principles of Heterogeneous Catalysis," pp. 458-459, Academic Press Inc., New York, 1967, for rate equations assembled in tabular form for various controlling mechanisms for the two reactions $A \rightleftharpoons B$ and $A + B \rightleftharpoons C$.

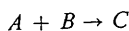
³Equations (9-32), (9-38), and (9-39) were developed for constant-temperature conditions. Hence the various specific rates k and equilibrium constants K are termed constants. They are, in theory, constant with respect to pressure and conversion but change with temperature.

nism studies for specific reactions has been summarized by Thomas and Thomas. Interesting comparisons of concepts and correlations for homogeneous and heterogeneous catalytic reactions are presented by Boudart.¹

The quantitative interpretation of kinetic data in terms of this type of rate equation is illustrated in Sec. 9-6. However, before we proceed to the evaluation of constants in rate equations, let us consider some of the implications of Eqs. (9-32), (9-38), and (9-39) from a qualitative standpoint.

The adsorption terms in the denominator are all different. For surface rate controlling, adsorption equilibrium groups ($K_i C_i$) are included for each component, and the entire term is squared because reaction is between adsorbed A and adsorbed B . For adsorption of A controlling, no group for the adsorption of A is present, and the entire term is to the first power. For desorption controlling, no group for product C is in the adsorption term. These differences result in separate relationships between the pressure and the rate. Hence the treatment of data as a function of pressure provides a useful method for distinguishing between equations. Temperature is not so useful a variable as pressure because all the constants are strong functions of temperature, and many of the same ones are part of each equation. The primary value of temperature is to determine the energies of activation for the adsorption and surface processes. Measurements at different conversion levels can be used to evaluate various equations, but in general it is difficult to draw valid conclusions about mechanisms. Quantitative treatment of conversion-vs-rate data is illustrated in Example 9-2. The importance of total pressure as a variable to study mechanism is shown in the following example.

Example 9-1 A solid-catalyzed gaseous reaction has the form



Sketch curves of the *initial* rate (rate at zero conversion) vs the total pressure for the following cases:

- The mechanism is the reaction between adsorbed A and adsorbed B molecules on the catalyst. The controlling step is the surface reaction.
- The mechanism is the same as (a), but adsorption of A is controlling.
- The mechanism is the same as (b), but desorption of C is controlling. Assume that the overall equilibrium constant is large with respect to the adsorption equilibrium constants.
- The mechanism is a reaction between adsorbed A and B in the gas phase. The controlling step is the surface reaction.

In each instance suppose that the reactants are present in an equimolar mixture.

¹Michel Boudart, "Kinetics of Chemical Processes," chaps. 8 and 9, Prentice-Hall, Inc., Englewood Cliffs, N.J., 1968.

Solution For a gaseous reaction the concentration C of any component i is proportional to its partial pressure; for an ideal-gas mixture $C_i = p_i/R_g T$. Hence at constant temperature partial pressures may be substituted for C_i in the rate equations, causing a change only in the value of the constants. At zero conversion the pressure of product C is zero, and for an equimolar mixture

$$p_A = p_B = \frac{1}{2}p_t$$

For the first three cases Eqs. (9-32), (9-38), and (9-39) are the appropriate rate equations.

(a) At initial conditions Eq. (9-32) simplifies to

$$r_0 = k_s \bar{C}_m K_A K_B \frac{\frac{1}{4} p_t^2 / (R_g T)^2}{[1 + \frac{1}{2}(K_A + K_B) p_t / R_g T]^2}$$

By combining constants we may write this expression as

$$r_0 = \frac{a p_t^2}{(1 + b p_t)^2} \quad (\text{A})$$

where a and b are the resulting overall constants.

(b) In a similar manner Eq. (9-38) for the adsorption of A controlling may be reduced to the form

$$r_0 = k_a \bar{C}_m \frac{\frac{1}{2} p_t / R_g T}{1 + \frac{1}{2} K_B p_t / R_g T} = \frac{a' p_t}{1 + b' p_t} \quad (\text{B})$$

(c) Equation (9-39), for the case where desorption of C is controlling, may be written

$$r_0 = k_d \bar{C}_m K \frac{\frac{1}{4} p_t^2 / (R_g T)^2}{1 + \frac{1}{2}(K_A + K_B) p_t / R_g T + \frac{1}{4} K_C K p_t^2 / (R_g T)^2}$$

If the equilibrium constant K is large with respect to K_A , K_B , and K_C , only the last term in the denominator is important, and the result is

$$r_0 = \frac{k_d \bar{C}_m}{K_C} = a'' \quad (\text{C})$$

Equation (A) for surface reaction controlling shows that the initial rate will be proportional to the square of the pressure at low pressures and will approach a constant value at high pressures. This type of relation is shown in Fig. 9-3a. The case for adsorption controlling is indicated in Fig. 9-3b, and that for desorption controlling is shown in Fig. 9-3c. If the equilibrium constant were not very large for case (c), the initial rate equation would be as shown in Fig. 9-3a.

(d) For this case the rate equation can be obtained from Eq. (9-18). Combining this with Eqs. (9-15) and (9-24) for the equilibrium values for $\bar{C}_A$ and $\bar{C}_C$ gives

$$r = k_s \bar{C}_v \left(K_A C_A C_B - \frac{K_C}{K_s} C_C \right)$$

Since there is no adsorption of B in this instance, Eq. (9-27) for $\bar{C}_v$ becomes

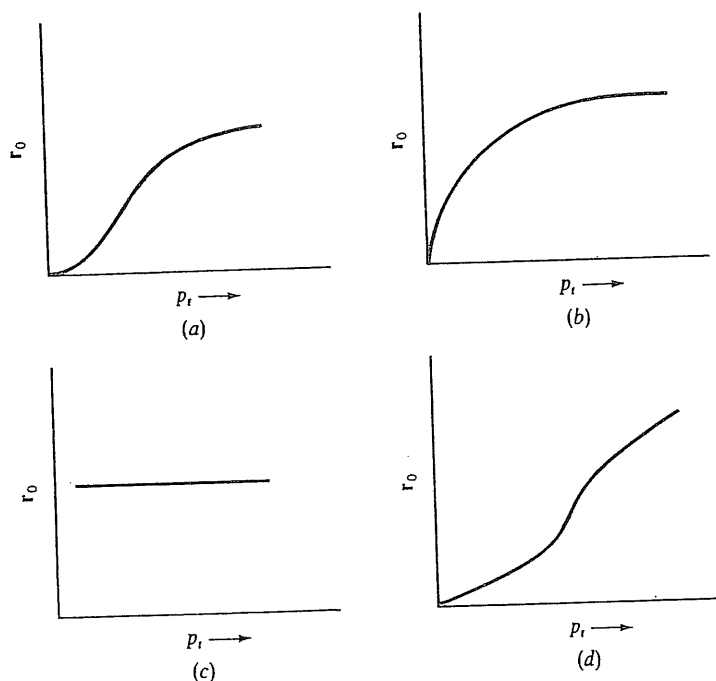


Fig. 9-3 Initial rate vs total pressure for reaction $A + B \rightarrow C$

$$\bar{C}_v = \frac{\bar{C}_m}{1 + K_A C_A + K_C C_C}$$

The relationship of the equilibrium constants [Eq. (9-31)] is

$$K = \frac{K_A K_s}{K_C}$$

Substituting these two expressions into the rate equation yields

$$r = k_s K_A \bar{C}_m \frac{C_A C_B - (1/K) C_C}{1 + K_A C_A + K_C C_C} \quad (D)$$

Equation (D) is the appropriate expression for the surface rate controlling, with no adsorption of B . At initial conditions, and replacing concentrations with partial pressures, it becomes

$$r_0 = k_s K_A \bar{C}_m \frac{\frac{1}{4} p_t^2 / (R_g T)^2}{1 + \frac{1}{2} K_A p_t / R_g T} = \frac{a''' p_t^2}{1 + b''' p_t} \quad (E)$$

A schematic diagram of Eq. (E) shows the rate proportional to p_t at high pressures (Fig. 9-3d).

Suppose that experimental rate data for the reaction $A + B \rightarrow C$ were obtained over a wide range of total pressures, all at the same temperature. Comparison of a plot of the observed results with curves such as shown in Fig. 9-3 would be of value in establishing the most accurate rate equation. However, it is sometimes difficult to cover a wide enough range of pressures to observe all the changes in shape of the curves.

9-6 Quantitative Interpretation of Kinetic Data

To evaluate the rate and adsorption equilibrium constants in equations such as (9-32) rate data are needed as a function of concentrations in the fluid phase. Data are required at a series of temperatures in order to establish the temperature dependency of these constants. The proper concentrations to employ are those directly adjacent to the site. In the treatment that follows we shall suppose that these local concentrations have been established from the measurable concentrations in the bulk stream by the methods to be given in Chaps. 10 and 11. Our objective here is to find the most appropriate rate equation at a catalyst site.

Statistical methods are required to obtain the best fit of the equation to the kinetic data. Minimizing the deviations between the observed rate and that predicted from the equation is straightforward as long as the constants are linearly related in the rate equation.¹ When nonlinearities exist the analysis is more complicated.² There are many illustrations of the method for evaluating the constants in equations such as (9-32) from experimental rate measurements.³ The sequence for this process is usually as follows:

1. Assume various mechanisms and controlling steps for each mechanism. Develop a rate equation for each combination of mechanism and controlling step.
2. Determine the numerical values of the constants which give the best fit of each equation to the observed rate data.
3. Choose the equation which best fits the data and agrees with available independent information about the reaction.

In the following simple example the evaluation of the constants is carried out for a single combination of mechanism and controlling step.

¹O. A. Hougen and K. M. Watson, "Chemical Process Principles," vol. III, "Kinetics and Catalysis," John Wiley & Sons, Inc., New York, 1948.

²G. E. P. Box and H. L. Lucas, *Biometrika*, **46**, 77 (1959); J. R. Kittrell, W. G. Hunter, and C. C. Watson, *AIChE J.*, **11**, 105 (1965), *AIChE J.*, **12**, 5 (1966); L. Lapidus and T. I. Peterson, *AIChE J.*, **11**, 891 (1965).

³J. R. Kittrell, Reiji Mezaki, and C. C. Watson, *Brit. Chem. Eng.*, **11**, 15 (1966), *Ind. Eng. Chem.*, **57**, 18 (1965); W. G. Hunter and Reiji Mezaki, *AIChE J.*, **10**, 315 (1964).

Example 9-2 Olson and Schuler¹ determined reaction rates for the oxidation of sulfur dioxide, using a packed bed of platinum-on-alumina catalyst pellets. A differential reactor was employed, and the partial pressures as measured from bulk-stream compositions were corrected to fluid-phase values at the catalyst surface by the methods described in Chap. 10 (see Example 10-1). The total pressure was about 790 mm Hg.

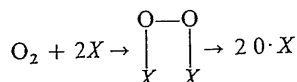
From previous studies² and the qualitative nature of the rate data a likely combination appeared to be a controlling surface reaction between adsorbed atomic oxygen and unadsorbed sulfur dioxide. In order to determine all the constants in the rate equation for this mechanism, it is necessary to vary each partial pressure independently in the experimental work. Thus measuring the rate of reaction at different total pressures but at constant composition is not sufficient to determine all the adsorption equilibrium constants. Similarly, if the data are obtained at constant composition of initial reactants but varying conversions, the partial pressures of the individual components do not vary independently. However, in these cases it is possible to verify the validity of the rate equation even though values of the separate adsorption equilibrium constants cannot be ascertained. Olson and Schuler studied the effect of conversion alone and obtained the data in Table 9-1 at 480°C.

Table 9-1

r, g moles/(hr)(g catalyst)	Partial pressure (atm) at catalyst surface		
	SO ₃	SO ₂	O ₂
0.02	0.0428	0.0255	0.186
0.04	0.0331	0.0352	0.190
0.06	0.0272	0.0409	0.193
0.08	0.0236	0.0443	0.195
0.10	0.0214	0.0464	0.196
0.12	0.0201	0.0476	0.197

Test the proposed mechanism by evaluating the constants in the rate equation and then comparing experimental rates of reaction with values computed from the equation.

Solution To develop an expression for the rate of reaction we must postulate the method of obtaining adsorbed atomic oxygen. If it is supposed that molecular oxygen is first adsorbed on a pair of vacant centers and that this product then dissociates into two adsorbed atoms, the process may be written



Since the surface reaction is controlling, the adsorption of oxygen must be at

¹ R. W. Olson, R. W. Schuler, and J. M. Smith, *Chem. Eng. Progr.*, **46**, 614 (1950).

² O. Uyehara and K. M. Watson, *Ind. Eng. Chem.*, **35**, 541 (1943).

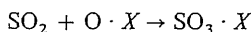
equilibrium. Then the concentration of adsorbed atomic oxygen is given by the equilibrium equation

$$K_O = \frac{\bar{C}_O^2}{p_{O_2} \bar{C}_v^2}$$

where $\bar{C}_O$ represents the concentration of adsorbed atomic oxygen and $\bar{C}_v$ represents the concentration of vacant centers. Solving this expression for $\bar{C}_O$ yields

$$\bar{C}_O = K_O^{\frac{1}{2}} p_{O_2}^{\frac{1}{2}} \bar{C}_v \quad (A)$$

The surface-reaction step is represented by



$$r_s = k_s \left(p_{SO_2} \bar{C}_O - \frac{1}{K_s} \bar{C}_{SO_3} \right) \quad (B)$$

The concentration of SO_3 adsorbed on the catalyst is given by the conventional equilibrium expression

$$\bar{C}_{SO_3} = K_{SO_3} p_{SO_3} \bar{C}_v \quad (C)$$

Substituting the values of $\bar{C}_O$ and $\bar{C}_{SO_3}$ in Eq. (B) gives

$$r_s = k_s \left(K_O^{\frac{1}{2}} p_{SO_2} p_{O_2}^{\frac{1}{2}} - \frac{K_{SO_3}}{K_s} p_{SO_3} \right) \bar{C}_v \quad (D)$$

The total concentration of centers is the summation

$$\bar{C}_m = \bar{C}_v + K_O^{\frac{1}{2}} p_{O_2}^{\frac{1}{2}} \bar{C}_v + K_{SO_3} p_{SO_3} \bar{C}_v + K_{N_2} p_{N_2} \bar{C}_v \quad (E)$$

The last term is included to take into account the possibility that N_2 may be adsorbed on the catalyst. Eliminating $\bar{C}_v$ from Eq. (D) by introducing Eq. (E) and noting that $K_O^{\frac{1}{2}} K_s / K_{SO_3} = K$, we may write Eq. (D) as

$$r = \frac{k_s \bar{C}_m K_O^{\frac{1}{2}} [p_{SO_2} p_{O_2}^{\frac{1}{2}} - (1/K) p_{SO_3}]}{1 + K_O^{\frac{1}{2}} p_{O_2}^{\frac{1}{2}} + K_{SO_3} p_{SO_3} + K_{N_2} p_{N_2}} \quad (F)$$

Because the partial pressures of sulfur dioxide and sulfur trioxide are both small, the value of p_{N_2} will not vary significantly with conversion. Hence $K_{N_2} p_{N_2}$ may be regarded as a constant in Eq. (F). Since conversion was the only variable causing the composition to change, p_{O_2} and p_{SO_3} and p_{SO_2} are not independent, but are related to the initial-constant composition and the conversion. Hence p_{O_2} in the denominator of Eq. (F) can be expressed in terms of conversion or, more conveniently, in terms of p_{SO_3} . Thus Eq. (F) may be simplified to

$$r = \frac{k_s \bar{C}_m K_O^{\frac{1}{2}} [p_{SO_2} p_{O_2}^{\frac{1}{2}} - (1/K) p_{SO_3}]}{A' + B' p_{SO_3}}$$

Combining the constants and rearranging leads to the form

$$A + B p_{SO_3} = \frac{p_{SO_2} p_{O_2}^{\frac{1}{2}} - (1/K) p_{SO_3}}{r} = R \quad (G)$$

Since conversion was the only variable, the rate equation contains only two constants, A and B . This means that individual adsorption equilibrium constants cannot be determined.

From the overall equilibrium constant K and the data for the partial pressures and rates, the right-hand side of Eq. (G), designated as R , can be evaluated for each rate. The equilibrium constant decreases with temperature, the reaction being exothermic. However, it is still about 73 at 480°C, as estimated from the expression

$$\ln K = \frac{22,200}{R_g T} - 10.5$$

Table 9-2

r , g moles/(hr)(g catalyst)	p_{SO_3} , atm	R	Rp_{SO_3}	$p_{\text{SO}_3}^2 \times 10^4$
0.02	0.0428	0.521	0.0223	18.3
0.04	0.0331	0.372	0.0123	11.0
0.06	0.0272	0.294	0.0081	7.40
0.08	0.0236	0.241	0.00569	5.57
0.10	0.0214	0.203	0.00433	4.58
0.12	0.0201	0.174	0.00349	4.03
Total	0.168	1.805	0.0562	50.8

Table 9-2 shows the values of R computed from the data, with $K = 73$. The most probable values of the constants are those corresponding to a least-mean-square fit of the data to a straight line of the general form

$$a + bx = y \quad (\text{H})$$

Comparing Eqs. (G) and (H), we have

$$y = R \quad x = p_{\text{SO}_3} \quad a = A \quad b = B$$

The values of a and b are given by the two expressions¹

$$b = \frac{\sum yx - \sum x \sum y / n}{\sum x^2 - (\sum x)^2 / n} \quad (\text{I})$$

$$a = \frac{\sum y - b \sum x}{n} \quad (\text{J})$$

where n is the number of measurements.

Equations (I) and (J) may be applied to the present case. Thus $yx = Rp_{\text{SO}_3}$, etc.

¹These equations are obtained by writing an expression for the sum of the squares of the deviations of the data points from the line of Eq. (H), differentiating this equation with respect to both a and b , setting the differentiated equations equal to zero to apply the minimum-deviation restriction, and then solving the two equations for the values of a and b .